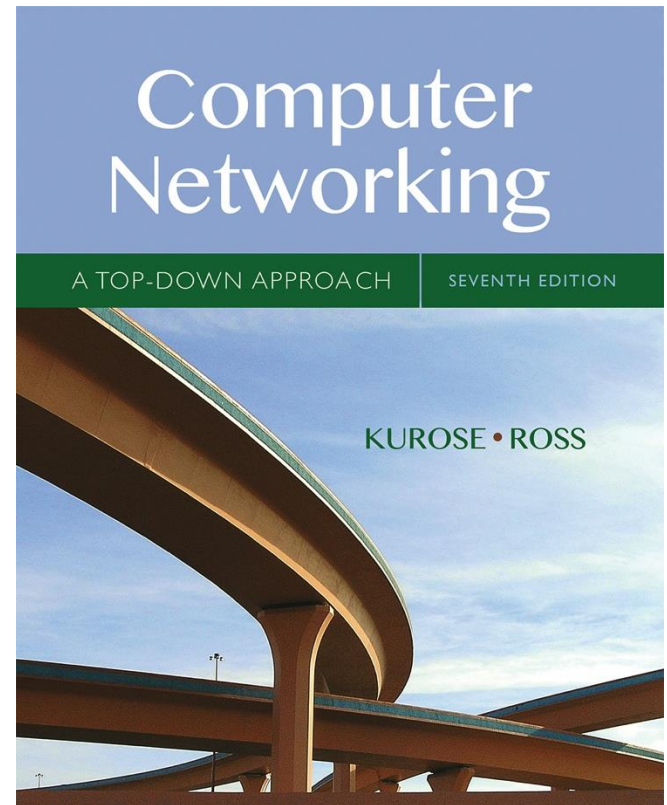


Chapter 9 Multimedia Networking



Computer Networking: A Top Down Approach

7th edition

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Multimedia networking: outline

9.1 multimedia networking applications

9.2 streaming *stored* video

9.3 voice-over-IP

9.4 protocols for *real-time*
conversational applications: RTP

9.5 network support for multimedia

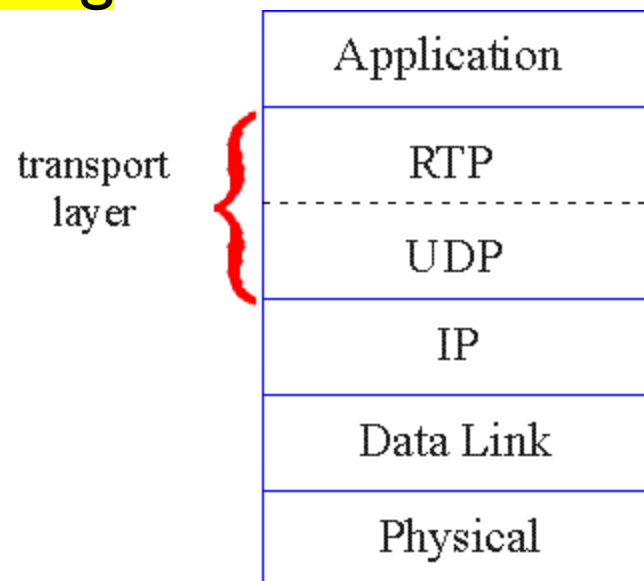
Real-Time Protocol (RTP)

- RTP specifies packet structure for packets carrying audio, video data
- RFC 3550
- RTP packet provides
 - payload type identification
 - packet sequence numbering
 - time stamping
- RTP runs in end systems
- RTP packets encapsulated in UDP segments
- interoperability: if two VoIP applications run RTP, they may be able to work together

RTP runs on top of UDP

RTP libraries provide transport-layer interface that extends UDP:

- port numbers, IP addresses
- payload type identification
- packet sequence numbering
- time-stamping



RTP example

example: sending 64 kbps PCM-encoded voice over RTP

- application collects encoded data in chunks, e.g., every 20 msec = 160 bytes in a chunk
- audio chunk + RTP header form RTP packet, which is encapsulated in UDP segment
- RTP header indicates type of audio encoding in each packet
 - sender can change encoding during conference
- RTP header also contains sequence numbers, timestamps

RTP and QoS

- RTP does *not* provide any mechanism to ensure timely data delivery or other QoS guarantees
- RTP encapsulation only seen at end systems (*not* by intermediate routers)
 - routers provide best-effort service, making no special effort to ensure that RTP packets arrive at destination in timely matter

RTP header

<i>payload type</i>	<i>sequence number</i>	<i>time stamp</i>	<i>Synchronization Source ID</i>	<i>Miscellaneous fields</i>
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payload type (7 bits): indicates type of encoding currently being used. If sender changes encoding during call, sender informs receiver via payload type field

Payload type 0: **PCM** mu-law, 64 kbps

Payload type 3: GSM, 13 kbps

Payload type 7: LPC, 2.4 kbps

Payload type 26: Motion JPEG

Payload type 31: H.261

Payload type 33: MPEG2 video

sequence # (16 bits): increment by one for each RTP packet sent

- ❖ **detect packet loss, restore packet sequence**

RTP header

<i>payload type</i>	<i>sequence number</i>	<i>time stamp</i>	<i>Synchronization Source ID</i>	<i>Miscellaneous fields</i>
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- ***timestamp field (32 bits long)***: sampling instant of first byte in this RTP data packet
 - for audio, timestamp clock increments by one for each sampling period (e.g., each 125 usecs for 8 KHz sampling clock)
 - if application generates chunks of 160 encoded samples, timestamp increases by 160 for each RTP packet when source is active. Timestamp clock continues to increase at constant rate when source is inactive.
- ***SSRC field (32 bits long)***: identifies source of RTP stream. Each stream in RTP session has distinct SSRC